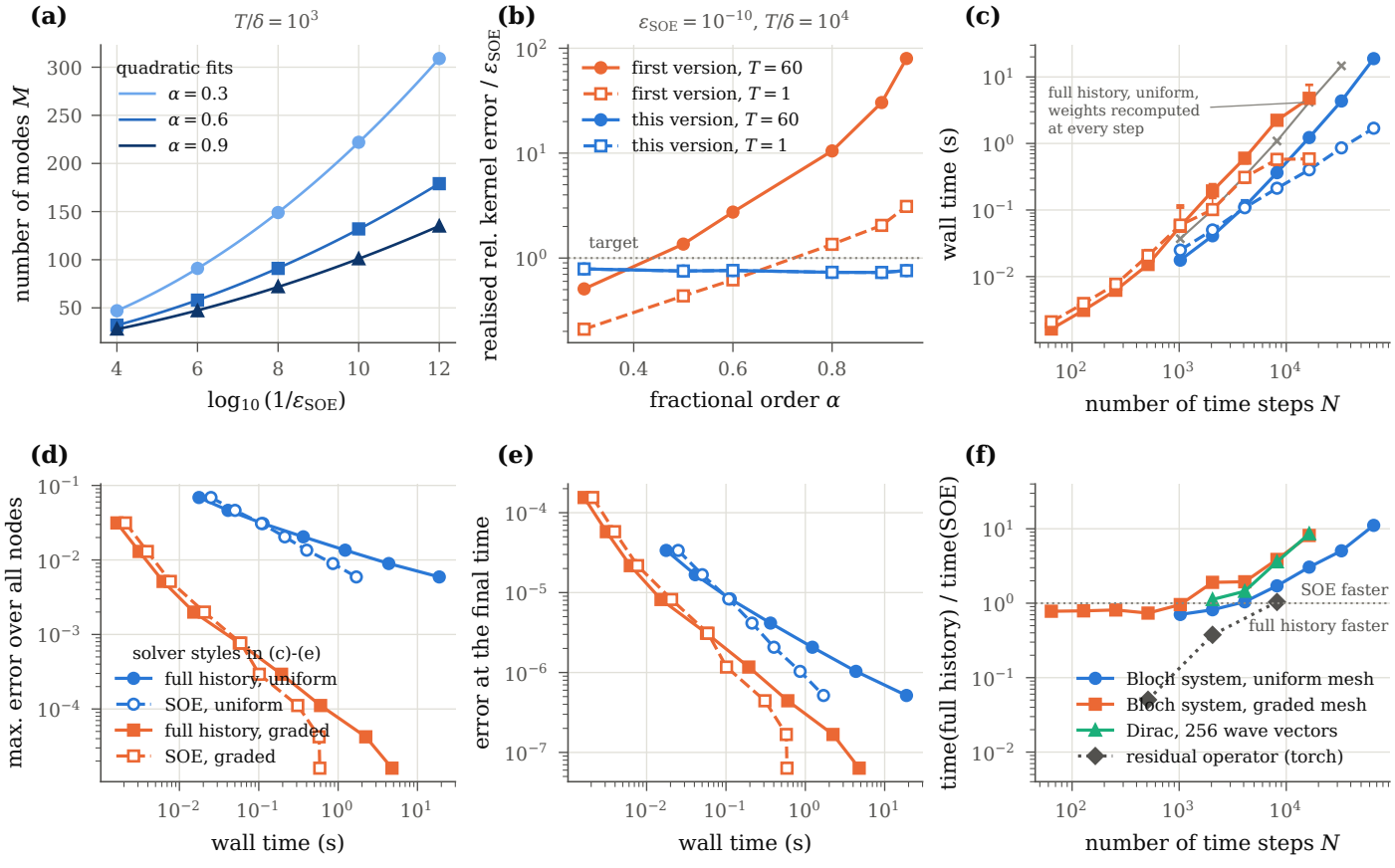
(a)
$T/\delta = 10^3$
quadratic fits
$\alpha = 0.3$
$\alpha = 0.6$
$\alpha = 0.9$
number of modes M
$\log_{10}(1/\varepsilon_{\mathrm{SOE}})$
(b)
$\varepsilon_{\mathrm{SOE}} = 10^{-10}$, $T/\delta = 10^4$
first version, $T = 60$
first version, $T = 1$
this version, $T = 60$
this version, $T = 1$
target
realised rel. kernel error / $\varepsilon_{\mathrm{SOE}}$
fractional order α
(c)
full history, uniform, weights recomputed at every step
wall time (s)
number of time steps N
(d)
solver styles in (c)-(e)
full history, uniform
SOE, uniform
full history, graded
SOE, graded
max. error over all nodes
wall time (s)
(e)
error at the final time
wall time (s)
(f)
SOE faster
full history faster
Bloch system, uniform mesh
Bloch system, graded mesh
Dirac, 256 wave vectors
residual operator (torch)
time(full history) / time(SOE)
number of time steps N